DATASCI 406 FA 26, Quiz 2.5

Name	Uniqname	UMID

Each question is worth 1 point. Circle one answer for each.

Question 1 (1 point)

Importance resampling draws $Y_1, \ldots, Y_m$ from g and resamples them with probabilities

$$\omega_i = \frac{f^*(Y_i)/g(Y_i)}{\sum_j f^*(Y_j)/g(Y_j)},$$

where $f = c\, f^*$ and c is unknown. Why does it still draw from f?

1. Using f in place of f^* multiplies the top and bottom of ω_i by c, so ω_i does not change.
2. Using f in place of f^* multiplies every ω_i by c, which would make $c\omega_i > 1$.
3. Using f would lead to bias. Using f^* corrects the bias.
4. It does not. c must be computed first, for example with `integrate()`.

Solution

Answer: 1.

Question 2 (1 point)

The chain $X(t) = 0.9\, X(t-1) + \sqrt{1 - 0.9^2}\, Z(t)$, with independent $Z(t) \sim N(0,1)$, has stationary distribution $N(0,1)$. It starts at $X(0) = 10$. Which statement is true?

1. $X(1) \sim N(0,1)$, because $N(0,1)$ is stationary.
2. The early $X(t)$ are not $N(0,1)$, so we drop them as burn in.
3. The draws are dependent, so their average cannot converge to $E(X) = 0$.
4. The chain never reaches $N(0,1)$, because it remembers its start.

Solution

Answer: 2. Stationarity says that *if* $X(t-1) \sim N(0,1)$ then $X(t) \sim N(0,1)$. With $X(0) = 10$, $E(X(1)) = 9$, so option 1 is false. Option 3 is wrong because MCMC averages converge for dependent draws when the chain has the properties in the video. Option 4 is wrong because each step keeps only 90% of the distance from 0, so the start is forgotten after about 40 steps.

Question 3 (1 point)

Consider a *symmetric Metropolis-Hastings algorithm* where $g(x \mid X^*) = g(X^* \mid x)$. At step t with previous value $X(t-1) = x$ it evaluates a proposal X^* using the expression:

$$U \le \frac{f^*(X^*)}{f^*(x)} \cdot \frac{g(x \mid X^*)}{g(X^* \mid x)} = \frac{f^*(X^*)}{f^*(x)}$$

where U is an independent uniform on $(0,1)$.

In one step, $f^*(X^*)/f^*(x) = 0.5$ and $U = 0.7$. What happens?

1. $X(t) = X^*$.
2. $X(t) = x$.
3. $X(t) = U$.
4. Nothing is recorded for time t, and a new proposal is drawn for $t+1$.

Solution

Answer: 2. $U = 0.7 > 0.5$, so the chain stays at x and records it again.

Question 4 (1 point)

A Gibbs sampler targets the bivariate normal with means 0, variances 1 and covariance 0.5, so $X_1 \mid X_2 = t \sim N(0.5\,t,\ 0.75)$. The current state is $(x_1, x_2) = (3, 2)$, and the next step updates x_1. From which distribution is the new x_1 drawn?

1. $N(0, 1)$ 2. $N(1, 0.75)$ 3. $N(1.5, 0.75)$ 4. $N(1, 1)$

Solution

Answer: 2. The full conditional of x_1 is evaluated at the current $x_2 = 2$, so the mean is $0.5 \times 2 = 1$. Option 1 is the marginal, option 3 conditions on the old x_1, and option 4 uses the marginal variance.

Rubric. 1 pt per question, no partial credit.